Convergence Tests for Infinite Series

Choosing among the direct comparison, integral, ratio, and alternating series tests

Directions

- Name the test you are using before you use it, and check its hypotheses.
- Give exact values (fractions, logarithms) rather than decimals.
- A convergence conclusion with no supporting limit or integral earns no credit — show the quantity the test depends on.

1. Consider $\displaystyle\sum_{n=0}^{\infty} \frac{1}{3^n}$.

(a) Find its exact sum.

(b) Use a direct comparison to explain why $\displaystyle\sum_{n=0}^{\infty} \frac{1}{3^n + n}$ also converges.

Answer: ________________

2. Apply the ratio test to $\displaystyle\sum_{n=1}^{\infty} \frac{n}{2^n}$. Simplify $\left|\dfrac{a_{n+1}}{a_n}\right|$, evaluate $L = \lim_{n\to\infty} \left|\dfrac{a_{n+1}}{a_n}\right|$, and state the conclusion.

Answer: ________________

3. Check that $f(x) = \dfrac{1}{x^2}$ is positive, continuous, and decreasing on $[1, \infty)$, then use the integral test on $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2}$. Evaluate $\displaystyle\int_1^{\infty} \frac{dx}{x^2}$ exactly and state what it proves.

Answer: ___________________

4. Let $S = \displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^3}$.

(a) Compute the fourth partial sum S_4 as an exact fraction.

(b) Give the alternating-series error bound for $|S - S_4|$ as an exact fraction.

Answer: ___________________

5. Use the ratio test on $\sum_{n=0}^{\infty} \frac{2^n}{n!}$. Simplify $\frac{a_{n+1}}{a_n}$ completely before taking the limit, then state the conclusion.

Answer: ________________

6. Consider $\sum_{n=1}^{\infty} \frac{1}{n^2 + 3n}$.

(a) Use a direct comparison with a p-series to show the series converges.

(b) Using $\frac{1}{n(n+3)} = \frac{1}{3}\left(\frac{1}{n} - \frac{1}{n+3}\right)$, telescope the partial sums and find the exact sum.

Answer: ________________

7. Use the integral test to decide whether $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ converges. Evaluate the improper integral exactly (a substitution $u = \ln x$ helps) and state the conclusion.

Answer: ____________________

8. For $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$, find the smallest number of terms N for which the alternating-series bound guarantees $|S - S_N| < 0.01$, and give the bound $|a_{N+1}|$ as an exact fraction.

Answer: ____________________

9. Use the ratio test to find the radius of convergence of the power series $\displaystyle\sum_{n=1}^{\infty} \frac{n\,(x-2)^n}{5^n}$, then state the open interval on which the ratio test guarantees convergence.

Answer: ________________

10. Show that $\displaystyle\sum_{n=1}^{\infty} (-1)^n \frac{n}{n^2+1}$ converges but is not absolutely convergent. Name the test used for each half of the argument, and verify the hypotheses of the test you use for convergence.

Answer: ________________